Shadi Rahimfallah

Ph.D. Candidate in Economics, York University
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PROFILE

Economics Ph.D. candidate at York University (expected 2027) with experience in economic modeling, econometrics, machine learning, and financial and market analysis. Research in macro-finance, firm productivity, electricity market design, and Treasury bond returns. Industry and consulting work in pricing, forecasting, and business cases, with a focus on large datasets and turning analysis into decisions for CEOs, shareholders, and client teams. Programming skills in Python, Stata, and SQL. Authorized to work in Canada.

EDUCATION

Ph.D. in Economics, York University, Toronto — 2022 – 2027 (expected)
Supervisors: Tasso Adamopoulos and Chaoran Chen

M.Sc. in Energy Economics, University of Tehran, Tehran — 2016 – 2019

B.Sc. in Theoretical Economics, Shahid Beheshti University, Tehran — 2012 – 2016

RESEARCH PAPERS

"Financial Frictions, Used Capital, and Misallocation", *Job Market Paper* [PDF] [SSRN] [Slides] [Code]
Under review, *Journal of Economic Dynamics and Control*

Dynamic model in which firms rent new and used capital, calibrated to Vietnamese firm data. Used capital lets credit-constrained firms operate closer to efficient scale and eliminates 39 percent of the productivity losses from borrowing constraints.

➤ Presented at: Canadian Economics Association Annual Meetings, 60th (2026) and 59th (2025); Macro Development Group, York University (2025, 2026)

"Second-Hand Machines, Firm Capability, and Technology Adoption", draft in progress

Firm surveys from fifteen countries and Vietnamese panel data: firms buying only used machines are smaller and less well managed, which explains most of their technology-adoption gap.

"Electricity Capacity Market", with Jeremy Lin (formerly PJM), work in progress

We study how rapidly growing large loads such as data centers affect PJM capacity market.

"News, the Economy, and Bond Returns", with Alireza Naeiji, work in progress

We study how macroeconomic data, together with news, predicts future Treasury excess returns, and which information each source contributes.

PROFESSIONAL EXPERIENCE

Research Assistant, York University — 2023 – 2025

Responsibilities: *Replicated a World Bank cluster-randomized land-titling trial in the Philippines from raw survey data (balance, attrition, ANCOVA intent-to-treat effects) and extended it to consumption risk and risk-sharing tests; built a country–year panel from World Bank and FAO data to study household exposure to a trade shock.*
Tools & Technologies: *Python, Stata.*

Senior Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis — 2019 – 2021

Responsibilities: *Led financial and market analysis for the CEOs and shareholders; built business cases for new tests, from market sizing and cost per test through pricing; led the case for COVID-19 PCR testing on existing equipment, which became a material contributor to revenue.*
Tools & Technologies: *Stata, SQL, Tableau.*

Financial & Economic Analyst | Accounting, Partogen Beh Tashkhis 2016 – 2019

Responsibilities: *Built demand, pricing, and market-sizing models from sales data; tracked competitor prices and market shares; developed the budgeting, forecasting, and financial reporting for company.*
Tools & Technologies: *Excel, SQL, Tableau.*

Research Assistant, University of Tehran 2018

Responsibilities: *Studied the design and regulation of Iran's day-ahead electricity market, analyzing how generator bidding and market power affect production and market efficiency; estimated two-way fixed-effects models of supply-chain disruptions and export credit coverage on export diversification, finding that disruptions bring temporary concentration and export credit aids diversification but not resilience.*
Tools & Technologies: *MATLAB, Stata, Python.*

Economic Analyst, Batab Kara, Economic and Investment Consulting Summer 2017

Responsibilities: *Wrote and presented economic and investment reports for client CEO and strategy teams; built a house-price forecasting model market-concentration risk.*
Tools & Technologies: *MATLAB, Stata.*

SKILLS

Software: Stata, Python (pandas, NumPy, statsmodels), MATLAB, R, SQL, Excel, Tableau, Git, LaTeX

Risk Modelling: Value-at-Risk, Expected Shortfall, stress testing and scenario analysis; credit risk

Financial Modelling: DCF/NPV and project appraisal, budgeting and forecasting, financial statement analysis

Econometrics & Data: Causal inference (RCTs, difference-in-differences), fixed-effects panels, time-series forecasting, machine learning; cleaning and linking large firm, survey, and market datasets

Industry Economics: Market sizing and pricing, demand elasticities, markup and market-power measurement

Economic Modelling: Structural general-equilibrium models built and calibrated from scratch; counterfactual and scenario analysis

Languages: English (fluent), Persian (native)

PUBLICATION

"Geoeconomics of Global Energy Transformation", with Farkhondeh Jabalameli, *Journal of World Sociopolitical Studies* 7(1): 55–87, 2023. [DOI]

TEACHING AND SERVICE

Teaching Assistant, *York University* 2022 – Present
Micro and Macro, Mathematics, Labour, Development Policy, Project Appraisal

Leader and Instructor, Graduate Econometrics Initiative, *University of Tehran* 2018 – 2020

Session Chair, *59th CEA Annual Meetings* 2025

Secretary, Economics Graduate Student Association, *York University* 2024

GRANTS & AWARDS

LA&PS Dissertation Fieldwork Fellowship, *York University* 2025

MacroFinance Summer School, full scholarship, *Princeton University* 2025

Computational Methods for Macroeconomics, full scholarship, *Econometric Society* 2023

Full Ph.D. Scholarship, *York University* 2022

Decision-Making under Risk, full scholarship, *University of Würzburg* 2021

M.Sc. and B.Sc. Full Scholarships (top 1% nationally), *Iran* 2016, 2012